

PRODUCT REQUIREMENTS DOCUMENT

PRISM

AI-Powered Automotive Procurement & Supplier Risk Management System

1. Product Overview

PRISM is an AI-powered procurement and supplier risk intelligence system for the automotive industry.

It connects:

- Production requirements
- BOM
- Inventory
- Supplier profiles
- Quotations
- Historical prices
- Supplier performance
- Delivery information

PRISM helps answer:

What to procure? → Who can provide it? → What could go wrong? → Why? → What should we do? → How can we negotiate better?

2. Problem Statement

Automotive procurement data is distributed across spreadsheets, ERP systems, emails and other sources.

Manual comparison and historical analysis take time, while emerging supplier risks may become visible only after affecting delivery or production.

Example:

Sept 10 → Sept 13 → Sept 17 → Sept 20

Repeated postponements may indicate emerging delivery and supplier-performance risk.

3. Product Vision

Transform supplier and procurement data into **early risk insights, evidence-based recommendations and better purchasing decisions.**

Reactive Procurement → Proactive Procurement Intelligence

4. Product Goals

- Identify emerging supplier risks early.
- Improve supplier evaluation and selection.
- Reduce manual quotation comparison.
- Provide historical price intelligence.
- Identify negotiation opportunities.
- Continuously monitor supplier conditions.
- Provide explainable AI recommendations.
- Maintain human control over important decisions.

5. Stakeholders

Stakeholder	Responsibility
Procurement Manager	Decisions, supplier evaluation, negotiation
Procurement Buyer/Analyst	RFQs, quotations, analysis
Supplier	Quotations, responses, delivery updates
Operations	Supplier verification and qualification
Finance/Approver	Cost and commercial approval
Production/Supply Chain	Production and component requirements

6. Scope In

PRISM covers:

- Automotive component procurement
- BOM, production and inventory-based planning
- Supplier and RFQ management
- Quotation processing and comparison
- Supplier intelligence and evaluation
- Historical price intelligence
- Supplier risk prediction and monitoring
- Decision and prescriptive recommendations

- Negotiation assistance
- Multi-agent procurement orchestration

7. Scope Out

PRISM does not cover:

- Fully autonomous purchasing
- Automatic supplier termination
- Direct financial transactions
- Physical factory or warehouse control
- Autonomous commercial agreement finalization

Pain Points & Product Response

Current Problem	PRISM Response
Scattered supplier information	Centralized intelligence
Manual quotation comparison	Structured comparison
Difficult historical analysis	Price intelligence
Late risk identification	Risk prediction & monitoring
Multiple signals difficult to combine	Multi-factor analysis
Risk causes unclear	Evidence-based explanation
Cheapest supplier may not be suitable	Multi-factor evaluation
Manual supplier follow-up	Continuous monitoring
Negotiation depends on experience	AI negotiation assistance
New suppliers lack history	Cold-start handling

Product Principle

Price + Delivery + Quality + Capacity + Lead Time + Performance + Risk + Commercial Conditions

8. Functional Requirements

FR1 — Authentication & RBAC

Secure login and role-based access.

FR2 — Procurement Dashboard

Requirements, suppliers, quotations, risks, KPIs and recommendations.

FR3 — Planning & Inventory

Analyze production, BOM, inventory and incoming supply.

FR4 — Supplier Intelligence

Evaluate supplier suitability using multiple factors.

FR5 — Risk Prediction

Assess delivery, shortage, capacity, quality, performance and commercial risks.

FR6 — Decision Support

Generate and evaluate procurement actions.

FR7 — Negotiation Assistance

Provide target prices, counteroffers and negotiation support.

FR8 — Monitoring

Monitor supplier, inventory, PO and delivery events.

9. Supplier Management

Supplier profile includes:

- Supplier ID and name
- Contact information
- Capacity
- Status
- Qualification
- Delivery history
- Quality history
- Performance information

10. RFQ Management

RFQ includes:

- Component
- Quantity
- Required date
- Specifications
- Selected suppliers

- RFQ status

RFQ Lifecycle

Created → Sent → Supplier Response → Quotation Received → Under Evaluation

11. Quotation Processing

Input Formats

- Excel / CSV
- API
- PDF

Extracted Information

- Supplier
- Component
- Quantity
- Unit price
- Currency
- MOQ
- Lead time
- Delivery date
- Payment terms
- Warranty
- Commercial conditions

Processing Flow

Upload → Processing/OCR → AI Extraction → Structured Data → Validation → Database

Validation Rule

Missing or ambiguous information is **flagged for validation, not fabricated.**

12. Supplier Intelligence

Suppliers are evaluated using:

Price | Delivery | Quality | Capacity | Lead Time | Performance | Status | Commercial Terms

13. Quotation Comparison

Dashboard compares:

- Price
- Quantity
- Delivery time
- Payment terms
- Warranty
- Commercial conditions
- Supplier performance
- Supplier score

Example

A: ₹125 | 5,000 units | 20 days

B: ₹130 | 5,000 units | 15 days

C: ₹122 | 4,000 units | 30 days

The system supports the procurement manager; it does not automatically make the final decision.

14. Historical Price Intelligence

Compare:

- Previous purchase price
- Current quotation
- Previous/current quantity
- Supplier
- Purchase date
- Price trends

Example

Previous ECU price: **₹120/unit**

Current quotation: **₹135/unit**

Difference: **₹15/unit**

This can indicate a negotiation opportunity after considering context.

15. Procurement KPIs

- Supplier Score
- Price Variance
- Historical Price Difference
- Potential Savings
- Delivery Performance
- Lead-Time Performance
- Inventory Coverage
- Risk Score
- Negotiation Opportunity

16. Risk Categories

1. Delivery Risk
2. Material/Stock Shortage
3. Capacity Risk
4. Quality Risk
5. Supplier Performance
6. Commercial/Price Risk

17. Risk Signals

- Delivery postponements
- Lead-time variability
- Quality rejection
- Quantity shortages
- Capacity constraints
- Supplier status changes
- Low inventory coverage
- Incoming supply delays
- Price changes
- Production pressure

Principle

One signal is not automatically proof of supplier failure.

PRISM evaluates multiple signals together with their context.

18. Risk Assessment

Each assessment contains:

Risk Type | Score | Severity | Evidence | Confidence | Impact

Example

Supplier: ABC Components

Risk: Delivery Risk

Severity: High

Evidence: Multiple postponements

Impact: Component shortage before production

Confidence: Medium

19. Decision & Prescriptive Workflow

Combines:

- Planning
- Supplier intelligence
- Risk information
- Historical data
- Procurement constraints

Possible actions:

- Continue supplier
- Negotiate earlier delivery
- Split order
- Select alternative supplier
- Increase safety stock

20. Negotiation Workflow

PRISM supports:

- Quotation analysis
- Historical price comparison

- Target price suggestions
- Counteroffer suggestions
- Negotiation strategy
- Message drafting
- Supplier response analysis

Example

Current quotation: ₹135/unit

Historical price: ₹120/unit

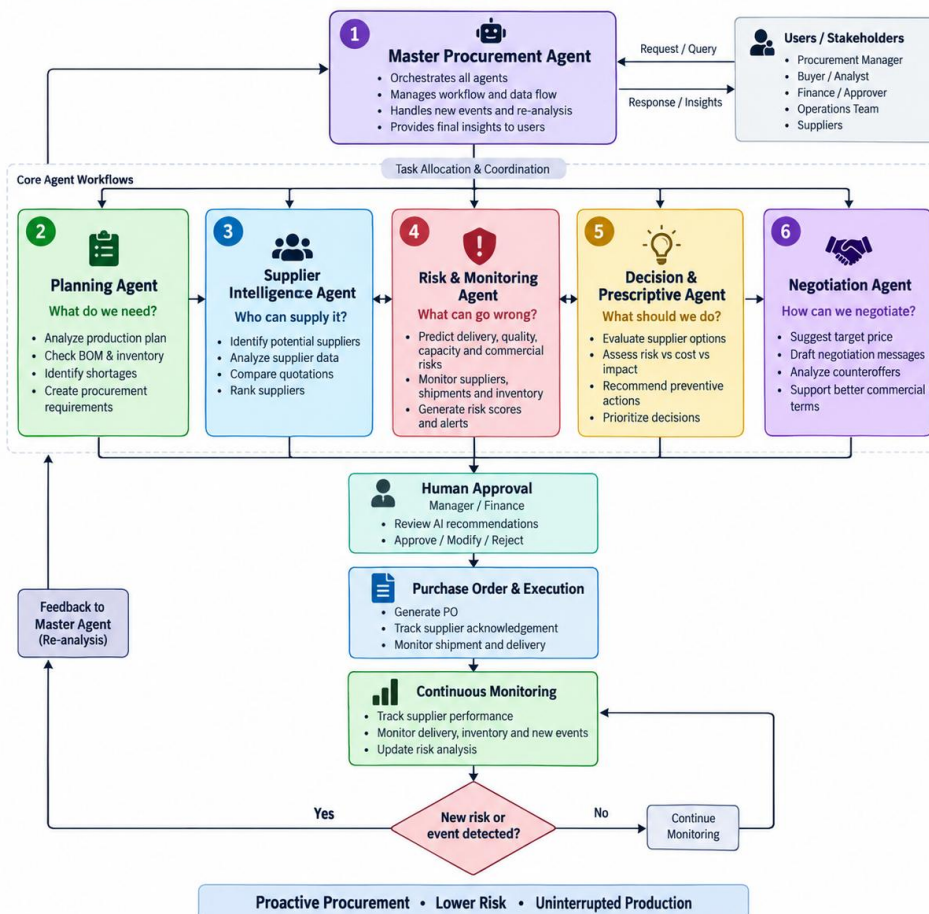
Possible target: ₹125/unit

These are decision-support suggestions reviewed by the procurement manager.

21. Multi-Agent Architecture

ProcurAgent – 6 Agent Workflow

AI-Powered Automotive Procurement & Supplier Risk Management



Master Procurement Workflow

User Request



Master Procurement Agent



Specialized Workflows

1. Planning & Inventory
2. Supplier Intelligence
3. Risk & Monitoring
4. Decision & Prescriptive
5. Negotiation



Aggregated Intelligence



Recommendation



Human Approval

The Master Agent coordinates information, aggregates outputs and triggers re-analysis when required.

22. New Supplier / Cold Start

New suppliers may have no historical performance.

PRISM does not fabricate history.

It evaluates available:

- Current quotation
- Quantity
- Promised delivery
- Lead time
- Capacity

- Qualification
- Market conditions
- Commercial terms

Output may include:

“Insufficient Historical Evidence”

or a lower-confidence assessment.

23. Human-in-the-Loop

AI can:

Analyze → Predict → Explain → Recommend → Assist Negotiation

Humans:

Review → Modify → Approve → Reject

24. Continuous Monitoring

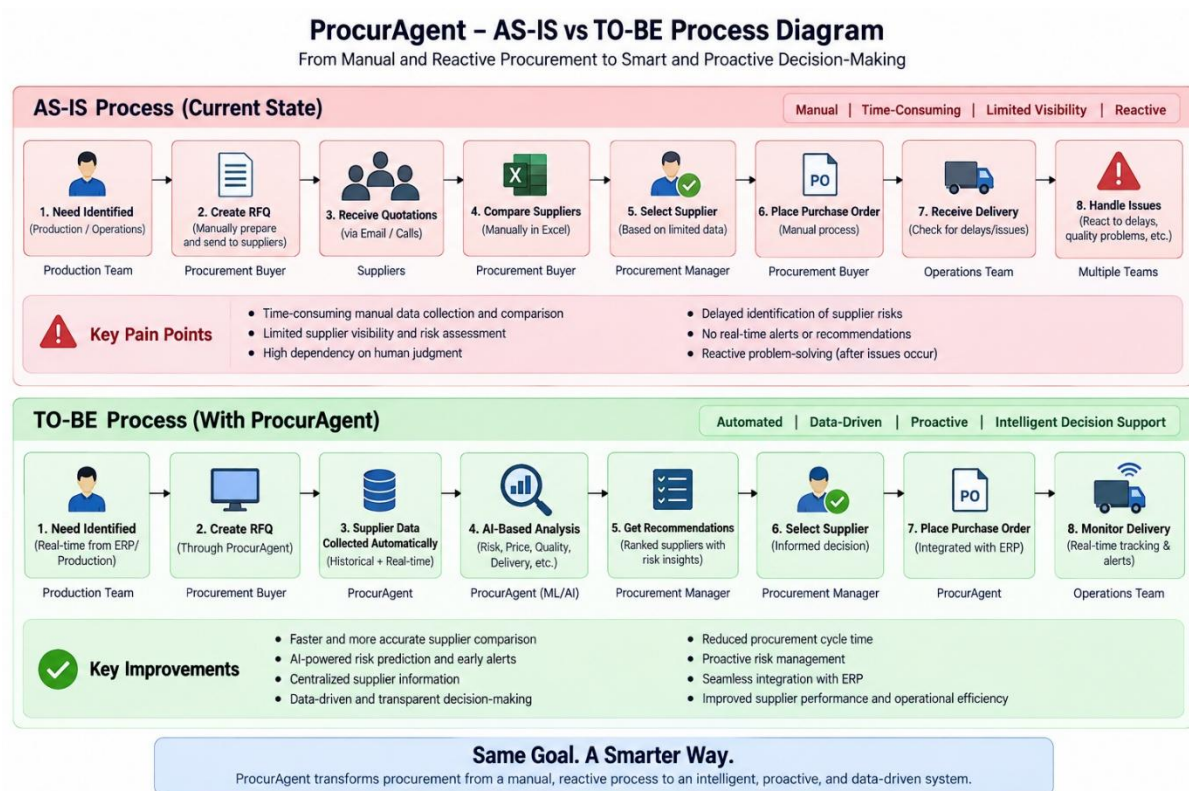
PRISM monitors:

- Supplier performance
- Inventory
- Purchase orders
- Shipment milestones
- Delivery milestones
- Supplier updates
- Reported issues

Re-analysis

New Event → Risk Analysis → Decision Analysis → New Recommendation

25. AS-IS → TO-BE

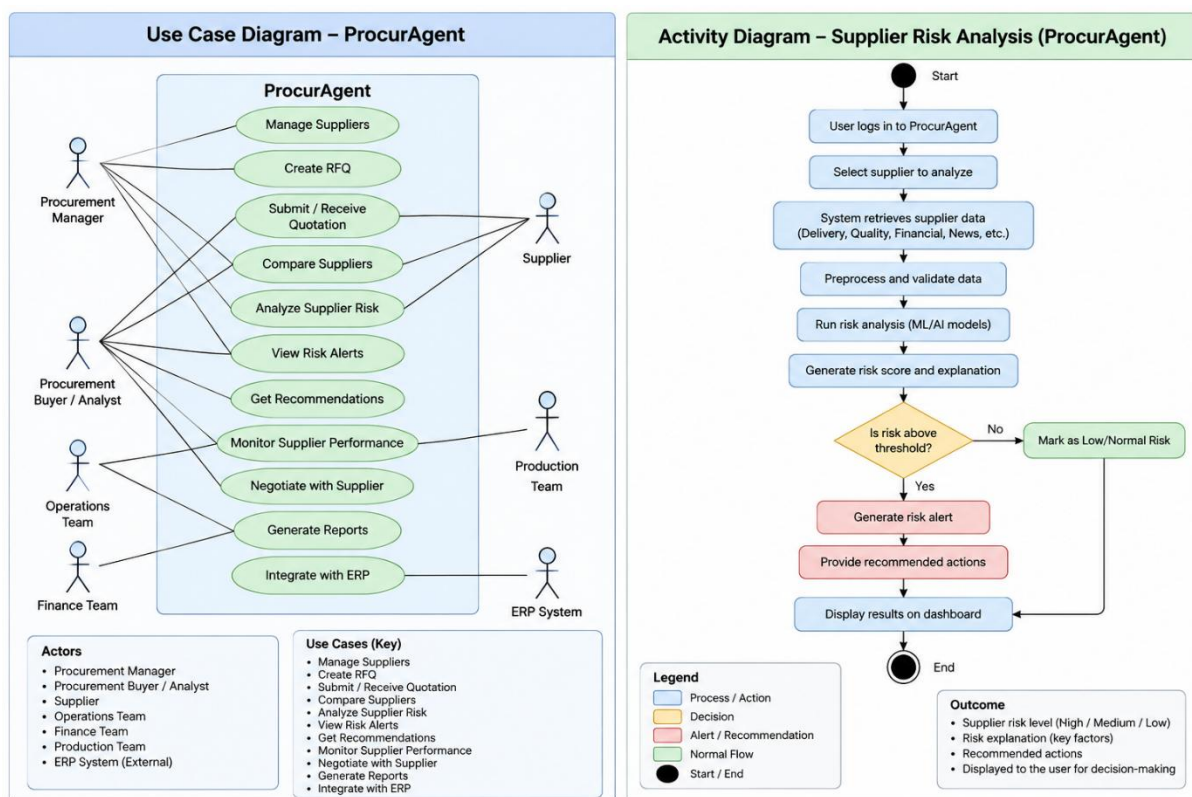


26. User Journey

Stage	User Action	System Output
Dashboard	Review KPIs	Procurement overview
Need Identification	Review requirements	Procurement requirement
RFQ	Create RFQ	RFQ sent
Quotation	Upload quotation	Structured quotation
Supplier Analysis	Compare suppliers	Supplier evaluation
Risk Analysis	Review risks	Risk assessment
Decision	Review alternatives	Recommendation
Negotiation	Review target/counteroffer	Negotiation support
Approval	Approve/modify/reject	Authorized decision
Monitoring	Track supplier	Alerts
Re-analysis	Respond to events	Updated recommendation

27. Key Use Cases

- Create RFQ
- Process quotation
- Compare suppliers
- Identify supplier risk
- Generate recommendation
- Support negotiation
- Monitor supplier performance



28. System Architecture

Users / External Systems

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React + Vite

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FastAPI API Gateway

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SNS Agent Workbench



Python ML / Intelligence



PostgreSQL

Responsibilities

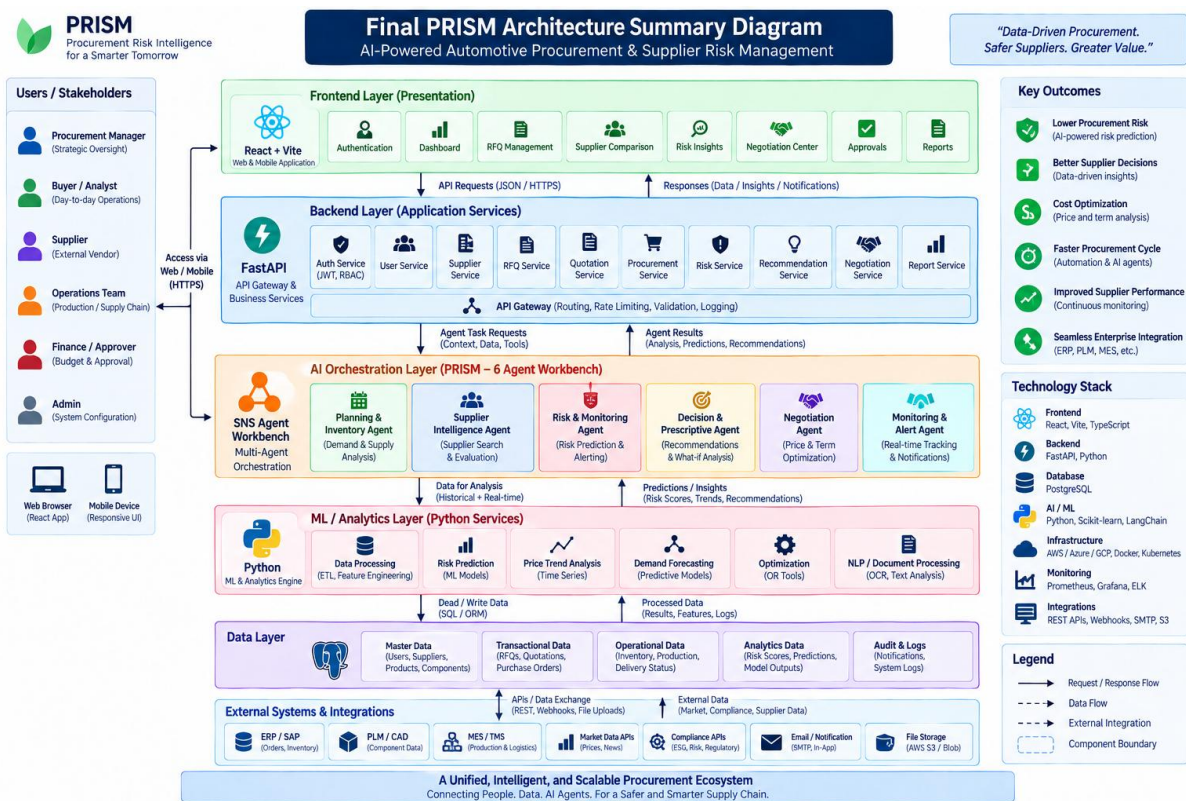
React: UI and dashboards

FastAPI: APIs, authentication, RBAC and integrations

SNS Workbench: Agent orchestration and reasoning

Python: ML and analytics

PostgreSQL: System of record



29. AI & ML

ML

- Delivery prediction
- Shortage prediction
- Quality-risk prediction
- Supplier-performance analysis
- Capacity assessment

- Price trend/anomaly analysis

AI Agents

- Workflow coordination
- Reasoning
- Risk explanation
- Recommendations
- Negotiation assistance
- Re-analysis

ML predicts patterns; AI agents reason about actions.

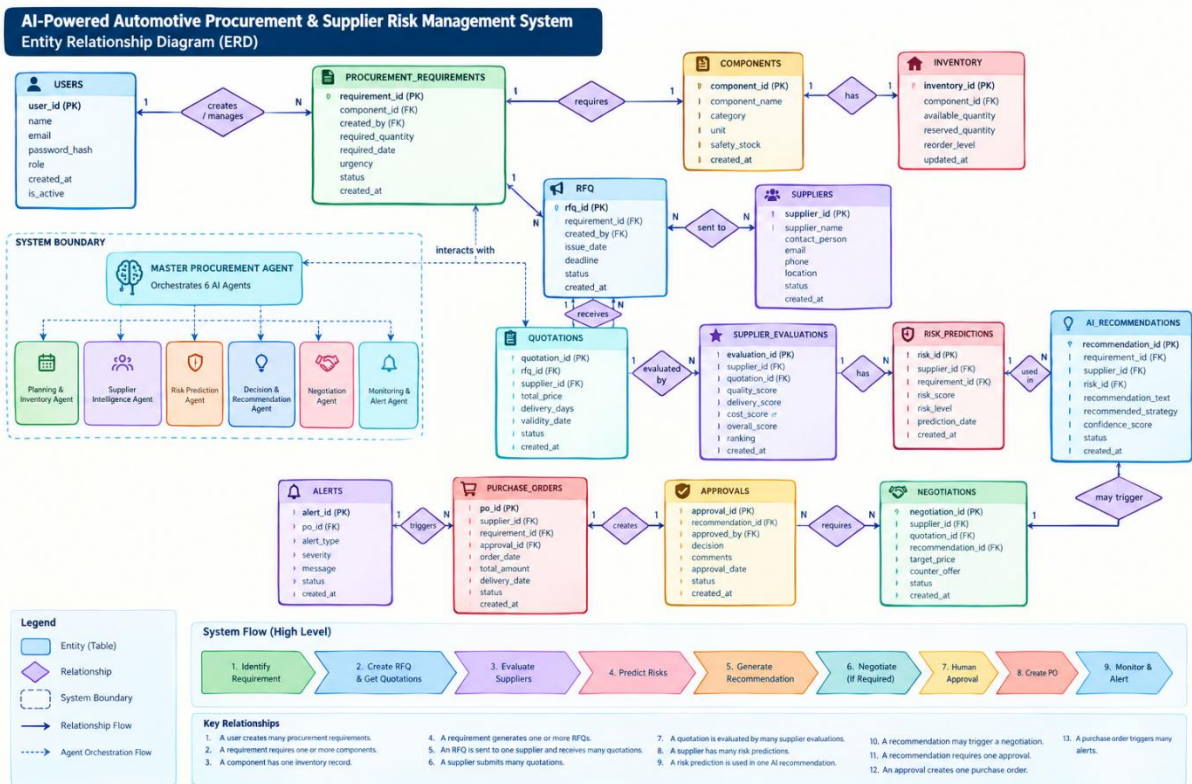
30. Data Requirements

- Customer orders
- Production plans
- BOM
- Component requirements
- Inventory
- Incoming supply
- Supplier profiles
- Supplier performance
- Delivery/quality history
- Supplier capacity
- Quotations
- Historical prices
- Market/reference information
- Shipment/delivery milestones
- Risk assessments
- Recommendations
- Approval records

Prototype data may use realistic synthetic automotive procurement data because real procurement data is generally confidential.

31. Major Database Entities

User | Supplier | Component | RFQ | Quotation | PO | Inventory | Supplier Performance
| Historical Purchase | Risk | Negotiation | Recommendation | Approval | Shipment



32. Core Business Rules

- Missing data must not be fabricated.
- New supplier history must not be fabricated.
- AI recommendations require evidence.
- AI must communicate uncertainty.
- Important commercial decisions require human review.
- Important decisions must be auditable.
- PostgreSQL is the source of truth.
- FastAPI is the central API gateway.
- Master Agent coordinates specialized workflows.

33. MoSCoW Prioritization

MUST HAVE

- Supplier & RFQ Management

- Quotation Processing
- Supplier Comparison
- Historical Price Comparison
- Risk Analysis
- Dashboard
- AI Recommendations
- Human Approval

SHOULD HAVE

- Real-time Monitoring
- Delivery Milestone Tracking
- Advanced Risk Prediction
- Advanced Price Intelligence

COULD HAVE

- Market Intelligence
- Scenario Simulation
- Advanced Optimization
- Additional Integrations

WON'T HAVE — CURRENT SCOPE

- Autonomous Purchasing
- Direct Financial Transactions
- Automatic Supplier Termination
- Physical Factory Control

ProcurAgent

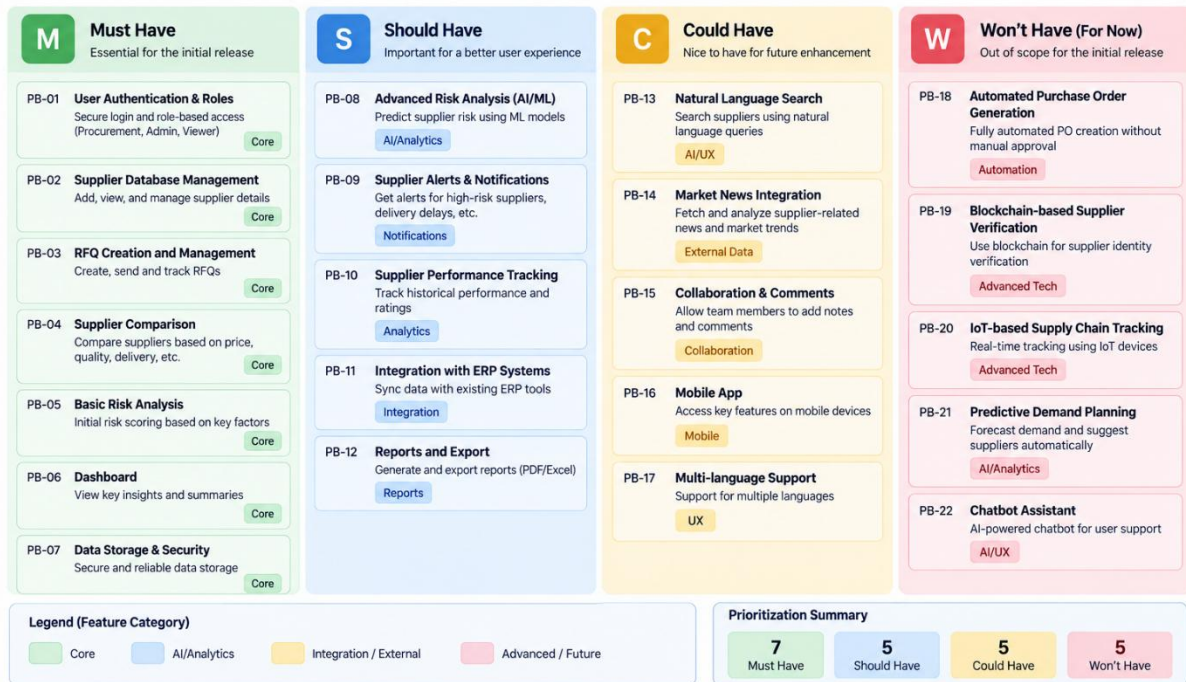
Product Backlog / MoSCoW Prioritization

Prioritizing features to build a smarter, safer and more efficient procurement system



Product Goal
Enable data-driven supplier selection and risk-aware procurement decisions.

*"Better Suppliers.
Lower Risks.
Brighter Outcomes."*



34. Success Criteria

PRISM should demonstrate that:

- Procurement requirements can be identified.
- Quotations can be structured and compared.
- Historical prices support benchmarking.
- Supplier performance can be evaluated.
- Emerging risks can be identified with evidence.
- AI can generate actionable recommendations.
- Negotiation suggestions can be generated.
- Humans can approve/reject recommendations.
- New suppliers are handled without fabricated history.
- Significant events can trigger re-analysis.

35. Differentiation

Traditional Procurement

Record → Compare → Purchase → Track

PRISM

Data → AI/ML Analysis → Risk Detection → Impact Analysis → Recommendation → Human Decision

PRISM acts as an **intelligent procurement co-pilot**.

36. Implementation Roadmap

Phase Focus

- 0 Project Setup
- 1 Database & FastAPI
- 2 Planning & BOM
- 3 Inventory & Procurement Requirements
- 4 RFQ & Quotations
- 5 SNS Agent Orchestration
- 6 Predictive ML
- 7 Prescriptive Decision Engine
- 8 AI Negotiation
- 9 Purchase Orders & Approvals
- 10 Monitoring & Feedback

Final Outcome

Earlier risk visibility + faster procurement analysis + better supplier evaluation + stronger negotiation preparation + explainable AI-assisted decisions.

PRISM

From reactive procurement to proactive procurement intelligence.

AI-POWERED AUTOMOTIVE PROCUREMENT & SUPPLIER RISK MANAGEMENT – FINAL ARCHITECTURE

